

Assignment 6: Estimation of Open Water Evaporation in Wadi Murwani Reservoir Using ECMWF ERA5

Due: Before the end of week 6

Instructions

Please submit your Overleaf-generated PDF, which should include any figures, your responses to the questions, and a link to your GitHub repository, via Blackboard before the specified due date. To do this, visit the course's Blackboard page, locate the assignment, attach the PDF, and then click 'Submit'. The necessary data and script for the course assignments can be found at <https://www.dropbox.com/scl/fo/wdqysdvvyxtrvb4y8euvf/h?rlkey=9f2pz8bsagpc2t8mm3ijhnfsy&dl=0> and should be downloaded to your hard drive (manually copy and paste the link into Chrome if clicking doesn't work). If you encounter any difficulties, first try to find solutions online, and then seek assistance from the instructor.

Background

Understanding and quantifying evaporation from reservoirs is critical, especially in arid regions like Saudi Arabia where water loss through evaporation can be substantial. In this assignment, we aim to estimate the potential evaporation in the Wadi Murwani Reservoir using the European Centre for Medium-Range Weather Forecasts (ECMWF) latest reanalysis, ERA5 (<https://doi.org/10.1002/qj.3803>). Reanalyses assimilate extensive situ and satellite observations into numerical weather prediction models to produce a temporally and spatially consistent, continuous record of the atmosphere, ocean, and land surface. These reanalyses are essential tools for various geo-environmental assessments.

Part 1: Data Download

1. Visit the Copernicus Climate Data Store (CDS) website (<https://cds.climate.copernicus.eu/>) and create a new account.
2. On the CDS website, search for "ERA5 hourly data on single levels from 1940 to present". This assignment only requires single-location data so select the first "ERA5 hourly time-series data on single levels from 1940 to present" and navigate to the "Download" tab.
3. CDS supports data downloads through both API requests and website interactions. Website interaction is straightforward and user-friendly, whereas API requests require an understanding of the download process and some coding skills. Because each request is subject to data size limits, downloading data for multiple years may not always be possible in a single request. In such cases, API requests are more efficient and flexible. Let's explore both methods.

- Website interaction:

In this tab, a form will appear where you need to select the following parameters:

- Variable: Total precipitation, 2m temperature
- Location: 22.25°N, 39.5°E
- Date: 2020-01-01 to 2025-12-31 (inclusive)
- Data Format: NetCDF

You must accept the terms and conditions of the dataset in the CDS. The option to view and accept these conditions is available at the end of the download form, just above the “Show API request” button. Click “Submit form” to submit your request, all of your recent requests are listed in “Your requests”, no matter they are submitted via website interaction or API request.

- API Request:

At the end of the download form, select “Show API request code”. This will display a block of Python code, which can be used as a template for further use. Based on that, you can create any request you need. Create a new Python script and paste the block.

4. The script uses the CDS Application Programming Interface (API) to download data. You can either specify the URL and key directly in the Python code or store them in the `.cdsapirc` configuration file in your home directory, such as `c:\Users\Arthur\`. The file should contain the following:

```
url: https://cds.climate.copernicus.eu/api
```

```
key: 67021:a1d00c40-8311-1b9d-b2de-960bae3aa239
```

Replace this key with your own CDS key (keep it as a secret), which can be found at <https://cds.climate.copernicus.eu/how-to-api>, or navigate to “Your profile” page to find it.

5. Run the script. Your request will be queued and the download will start after some time. Wait for the netCDF to be downloaded. If the wait is too long, use the data from the ERA5_–Data subfolder in the course materials, which was obtained using the same process.

Part 2: Data Pre-Processing

1. Create a new Python script named `assignment_6.py` in your `geo_env` repository.
2. Load the dataset using the `open_dataset` function from the `xarray` module. Be sure to specify the correct file path for the dataset:


```
import xarray
dset = xarray.open_dataset(r'C:\Downloads\ERA5_Data\download.nc')
```
3. Extract the relevant variables from the dataset, including air temperature (`t2m`), precipitation (`tp`), latitude, longitude, and time. Then, convert these variables into numpy arrays for further processing:

```
import numpy as np
t2m = np.array(dset.variables['t2m'])
tp = np.array(dset.variables['tp'])
latitude = np.array(dset.variables['latitude'])
longitude = np.array(dset.variables['longitude'])
time_dt = np.array(dset.variables['time'])
```

4. Convert the air temperature from K to °C and precipitation from m h^{-1} to mm h^{-1} for more intuitive interpretation:

```
t2m = t2m - 273.15
tp = tp * 1000
```

5. If the ERA5 dataset has four dimensions, indicating the presence of both final and preliminary data, compute the mean across the second dimension to simplify the dataset:

```
if t2m.ndim == 4:
    t2m = np.nanmean(t2m, axis=1)
    tp = np.nanmean(tp, axis=1)
```

6. Create a Pandas dataframe containing time series data for both air temperature and precipitation:

```
df_era5 = pd.DataFrame(index=time_dt)
df_era5['t2m'] = t2m
df_era5['tp'] = tp
```

7. Finally, plot the time series with the following commands:

```
df_era5.plot()
plt.show()
```

8. What is the average annual precipitation in mm y^{-1} ? Resample the data to annual time step and calculate the mean precipitation as follows:

```
annual_precip = df_era5['tp'].resample('YE').mean()*24*365.25
mean_annual_precip = np.nanmean(annual_precip)
```

9. Do you notice any trends in the time series for air temperature and/or precipitation?
10. How reliable do think the data on precipitation and air temperature are? Discuss whether one might be more reliable than the other.

Part 3: Calculation of Potential Evaporation (PE)

1. A variety of methods exist for calculating Potential Evaporation (PE). Some, like the Penman-Monteith method (<https://www.fao.org/3/X0490E/x0490e06.htm>), are more complex and require inputs for several meteorological variables. Others, such as the Hargreaves and Samani (1985) method (<http://dx.doi.org/10.13031/2013.26773>),

are simpler and only need air temperature. The equation for the Hargreaves and Samani (1985) method is:

$$PE = 0.0023 \times R_a \times \sqrt{T_{\max} - T_{\min}} \times (T_{\text{mean}} + 17.8), \quad (3)$$

where PE is the potential evaporation (mm d^{-1}), R_a is the extraterrestrial radiation (mm d^{-1}), and $T_{\max}$, $T_{\min}$, and T_{mean} are the maximum, minimum, and mean daily temperatures, respectively (all in $^{\circ}\text{C}$). To compute PE, use the `hargreaves_samani_1982` function from `tools.py`. This function requires $T_{\max}$, $T_{\min}$, T_{mean} , latitude, and day of the year as input. R_a is calculated internally from latitude and day of the year. Derive all inputs for the function from the hourly ERA5 data:

```
tmin = df_era5['t2m'].resample('D').min().values
tmax = df_era5['t2m'].resample('D').max().values
tmean = df_era5['t2m'].resample('D').mean().values
lat = 21.25
doy = df_era5['t2m'].resample('D').mean().index.dayofyear
```

2. Compute the PE using:

```
import tools
pe = tools.hargreaves_samani_1982(tmin, tmax, tmean, lat, doy)
```

3. Plot the PE time series:

```
ts_index = df_era5['t2m'].resample('D').mean().index
plt.figure()
plt.plot(ts_index, pe, label='Potential Evaporation')
plt.xlabel('Time')
plt.ylabel('Potential evaporation (mm d-1)')
plt.show()
```

4. What is the mean annual PE in mm y^{-1} ?
5. Based on the mean annual PE, what is the volume of water potentially lost from the reservoir through evaporation annually? Assume the reservoir covers an area of 1.6 km^2 , as determined from satellite imagery from December 12, 2023. Note that the PE calculated using the Hargreaves and Samani (1985) method is representative of a grass reference surface. To adjust this PE for open water surfaces like reservoirs, a surface coefficient, often called a crop coefficient in agricultural contexts, should be applied. However, for the purposes of this assignment, we will ignore this to keep things relatively simple.
6. Do you think evaporation from open water is generally higher or lower than evaporation from a grass surface?
7. Do you consider 1.6 km^2 to be a good estimate for the average area of the reservoir? Use the EO Browser (<https://apps.sentinel-hub.com/eo-browser/>) to answer this question.

8. Commit and push your code to the `geo_env` repository and create a concise PDF using Overleaf that incorporates all the figures and responses to the questions from this assignment, along with a link to your GitHub repository.